

Effortless In-Vehicle Task-Oriented Dialogue: Enhancing Natural and Efficient Interactions

Huifang Du*

College of Design and Innovation
Tongji University
Shanghai, China
duhuifang@tongji.edu.cn

Shuqin Li*

School of Computer Science
Hangzhou Dianzi University Information Engineering
School
Hangzhou, Zhejiang, China
shuqinlee9683@gmail.com

Yi Dai

Shanghai Research Institute for Intelligent Autonomous
Systems
Tongji University
Shanghai, China
2210999@tongji.edu.cn

Haofen Wang[†]

College of Design and Innovation
Tongji University
Shanghai, China
nobot@tongji.edu.cn

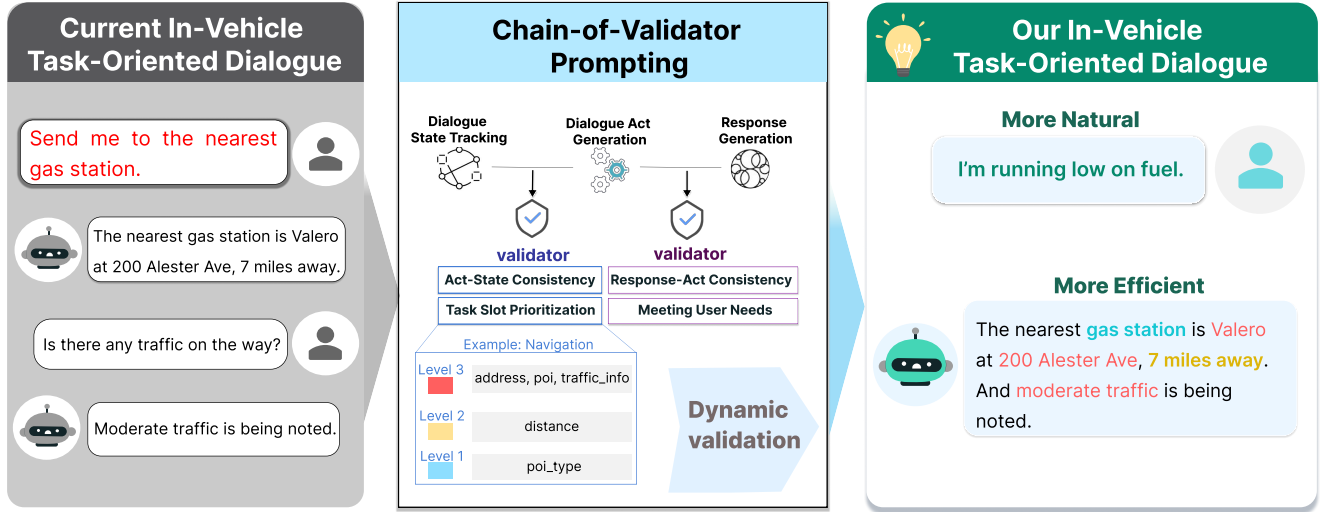


Figure 1: Tired of complex commands? Our human-centered in-vehicle TOD system enables natural interactions and prioritizes key info, boosting efficiency and safety.

Abstract

Have you ever struggled to give commands to your in-vehicle assistant? Have you stopped using it because of the lengthy, multi-turn dialogues? In-vehicle Task-Oriented Dialogue (TOD) systems often require detailed instructions and multiple interaction turns, increasing cognitive load, distracting drivers, and causing frustration.

*These authors contributed equally to this work.

[†]Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719799>

To address this, we propose a human-centered, efficient approach to in-vehicle TOD systems. We introduce a novel dataset covering 10 core in-vehicle scenarios, allowing users to express needs naturally without system-specific commands. Task slots with varying importance levels are defined to prioritize critical information and minimize unnecessary dialogue turns. We further propose a “chain-of-validator prompting” framework that validates and refines system responses in real time, reducing misunderstandings and improving efficiency. Experimental results demonstrate that our method surpasses conventional prompting in task completion, while subjective evaluations show superior appropriateness, usability, efficiency, naturalness, and reduced cognitive load.

CCS Concepts

• **Human-centered computing** → *HCI theory, concepts and models*; • **Computing methodologies** → **Natural language processing**.

Keywords

In-vehicle task-oriented dialogue, Human-centered conversation, Dialogue efficiency, Natural interaction

ACM Reference Format:

Huifang Du, Shuqin Li, Yi Dai, and Haofen Wang. 2025. Effortless In-Vehicle Task-Oriented Dialogue: Enhancing Natural and Efficient Interactions. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3706599.3719799>

1 Introduction

Task-oriented dialogue systems (TODS) have become an essential component in modern in-vehicle environments, assisting users with navigation, media control, and other context-specific tasks [6, 8]. Despite their utility, traditional TODS often rely on rigid, system-driven interaction paradigms that demand users provide detailed instructions through multiple turns of dialogue [41]. This approach not only increases cognitive load but also introduces significant distractions, raising concerns about safety and usability during driving [2, 9].

In response to these challenges, we propose a novel, human-centered methodology for in-vehicle TODS, designed to enhance naturalness and efficiency while minimizing distractions. Central to our approach is the development of a comprehensive dataset encompassing 10 core in-vehicle scenarios, each reflecting realistic user needs and behaviors. Unlike prior datasets, which often emphasize predefined system-specific commands, our dataset allows users to articulate their needs and observations naturally. For each scenario, we define task slots with varying importance levels, enabling the system to prioritize critical information and streamline interactions. By doing so, our approach reduces unnecessary dialogue turns and ensures that the system focuses on the most relevant aspects of user requests. Building on this foundation, we introduce a chain-of-validator prompting (CoVP) framework—a dynamic mechanism that leverages the dataset schema to validate and refine system responses in real time. This framework ensures greater interpretability and robustness by dynamically assessing system outputs against user inputs and contextual information [7, 13]. As a result, it can mitigate misunderstandings, facilitate faster task completion, and significantly enhance user satisfaction. Our work’s motivation and main contributions are shown in Figure 1.

Our experiments demonstrate the effectiveness of the proposed system through comparative evaluations against the conventional prompting method in task completion rate [33]. Results indicate substantial improvements in user perceptions like appropriateness, usability, efficiency, naturalness, and lower cognitive load.

We contribute to the field by presenting a novel dataset, an innovative dialogue framework, and a set of experimental insights that collectively advance the state-of-the-art in in-vehicle TODS. Through this work, we aim to pave the way for safer, more efficient,

and user-centric dialogue systems tailored to the unique challenges of the in-vehicle environment.

2 Related Work

2.1 Task-Oriented Dialogue Systems

Early TOD systems in the 1970s relied on manually crafted rules, which lacked scalability [36]. With the rise of deep learning, TOD systems shifted to data-driven approaches, improving multi-turn dialogue handling. Recent advancements focus on modular and end-to-end methods. Modular systems, involving sequential processes like DST [23], DPL [20, 30, 42], and RG [26, 29], face challenges such as error propagation [43]. End-to-end approaches, exemplified by SimpleTOD, SOLOIST, and UBAR [17, 25, 40], streamline architectures and improve adaptability. LLMs like GPT-4 have further enhanced performance [22, 39], and frameworks like DARD excel in multi-domain tasks [15]. However, TOD systems still struggle with ambiguities and cognitive load in complex scenarios. Our chain-of-validator prompting framework addresses these issues by ensuring real-time validation and refinement of responses.

2.2 Datasets for Task-Oriented Dialogue

Datasets are crucial for advancing TOD systems, covering diverse domains and complex scenarios. MultiWOZ supports cross-domain dialogues [2], while the In-car Assistant dataset features multi-turn, multi-domain interactions [10]. Schema-Guided Dialogue (SGD) spans 16 domains, enabling dynamic task switching [31], and TaskMaster combines real and scripted dialogues for scenarios like ticket booking [3]. SMCaFlow provides fine-grained data for calendar management [1], and DialoGLUE supports multi-task learning [24]. MetaLWOZ evaluates zero-shot and few-shot learning across 47 domains [21], while DuConv addresses Chinese knowledge-driven dialogues [37]. Despite their strengths, existing datasets focus on traditional tasks like reservations and lack consideration for in-vehicle environments and user cognitive load. To address this, we introduce a new dataset covering 10 core in-vehicle scenarios, incorporating task prioritization to optimize interactions for efficiency and naturalness.

2.3 Interaction Design for In-Vehicle Dialogue Systems

In-vehicle dialogue systems enhance the driving experience by focusing on functionality, user satisfaction, and seamless interaction. Context-awareness is key, enabling systems to adjust settings like cabin temperature or suggest rest areas, reducing user input, and improving naturalness [11, 14]. Systems can also predict needs, such as suggesting breaks or playing energizing music when detecting fatigue, boosting user experience [6]. Trust is built through reliable, predictable behavior, emphasizing stability, consistency, and minimizing unnecessary prompts to reduce cognitive load [5, 12, 34]. Participation mechanisms further enhance engagement and satisfaction by giving drivers more control [8]. We propose a user-centered approach, leveraging a tailored dataset and validation framework to reduce cognitive load, improve interaction efficiency, and enhance safety and naturalness in the driving experience.

3 Data Collection

To tackle cognitive load and dataset gaps in in-vehicle TOD systems, we introduce a user-centered dialogue dataset spanning 10 driving scenarios. It reduces system-specific instructions and multi-turn dialogues, promoting safety and efficiency via natural interactions. Sourced from MultiWOZ 2.0 [2], the In-car Assistant dataset [10], and our constructed in-vehicle dialogues, the dataset underwent a uniform processing pipeline to ensure quality and relevance to in-vehicle contexts. Processing steps are outlined below.

3.1 Data Processing

We began by filtering and selecting domains relevant to in-vehicle scenarios: **MultiWOZ Dataset**: As one of the most widely used TOD datasets, MultiWOZ covers seven domains, including hotel booking and restaurant reservation. We removed domains irrelevant to in-vehicle scenarios (e.g., taxi, train, and bus) and retained four domains: hotel, restaurant, attraction, and hospital. **In-car Assistant Dataset**: This dataset focuses on in-vehicle scenarios, containing 3,031 dialogues across three domains: calendar scheduling, weather information retrieval, and point-of-interest navigation. **Complementary Data**: To further enrich the dataset, we constructed dialogues for three additional in-vehicle domains: personal information, music, and radio. These domains cover tasks such as personal knowledge graph construction, music information retrieval, and radio channel navigation.

3.2 Optimizing Slots and Dialogues

For each domain, we constructed a schema and categorized task slots into three priority levels: high (level-3), medium (level-2), and low (level-1). This prioritization aims to optimize dialogue efficiency by ensuring the system focuses on critical information first. To ensure the reliability of the slot prioritization, we engaged three PhD students specializing in dialogue systems to review and validate the categorization. Figure 2 shows an example of task slots for the “find restaurant” task, categorized by importance levels. Using the constructed schema, we optimized the original dialogues by prompting the LLM GPT-4o. The optimization process adhered to two key principles: **Natural Expression**: User instructions were transformed into natural expressions of needs, feelings, or observations to enhance dialogue naturalness and user experience. **Efficient Interaction**: The system was designed to cover all slot information in the current priority level in each dialogue turn, reducing unnecessary interactions and improving efficiency. The optimization prompt for dialogues in MultiWOZ and In-car Assistant Dataset is shown in Figure 3. During the creation of our dataset, we replaced “refine” (red) with “create” and removed “Original Dialogue” (red) in the prompt.

3.3 Human Verification

All dialogues were human-verified for task completion, naturalness, and slot coverage. We built a high-quality in-vehicle dataset: 1,455 dialogues from MultiWOZ, 2,711 from the In-car Assistant dataset, and 274 newly constructed dialogues covering personal information, music, and radio. The final dataset spans 10 core scenarios, providing a robust research foundation. Table 1 shows slot priority processing reduced the original average turn number, with

```
{
  "name": "find_restaurant",
  "description": "search for places to wine and dine",
  "level-3": [
    "restaurant-name",
    "restaurant-area",
    "restaurant-food",
    "restaurant-address"
  ],
  "level-2": [
    "restaurant-pricerange"
  ],
  "level-1": [
    "restaurant-phone",
    "restaurant-postcode"
  ]
}
```

Figure 2: An example of task slots with varying importance in the “find restaurant” task.

Please **refine** the dialogue between the user and the in-vehicle task-oriented dialogue system to make the conversations more ****natural**** and ****efficient****. Follow the guidelines below:

Task Objectives

1. Transform User Instructions:
 - Convert the user's input into natural expressions of their needs, feelings, or observations to enhance conversational fluency.
2. Schema Structure and Intent Handling:
 - The provided "Schema" includes all intents and task slots in the dialogue. The slots are organized into three priority levels:
 - Level-3: Crucial for completing the task.
 - Level-2: Helpful but not critical.
 - Level-1: Less important; the task can proceed without this information.
 - Steps to follow:
 1. Meet the user's current needs based on their input.
 2. Optimize the system's responses to clarify all slots at the current priority level, progressing in the following order: ****Level-3 → Level-2 → Level-1****.
3. Output Format:
 - Present the refined dialogue in a ****list format**** that mirrors the structure of the original dialogue.

Input Details

- Original Dialogue: **{{}}**
- Schema: **{{}}**

Figure 3: Illustration of prompt optimization for conversations in MultiWOZ and In-car Assistant Dataset.

Dataset	Org. Avg. turns	Now Avg. turns
MultiWOZ	13.46	4.08
In-car Assistant	2.63	2.16
Complementary Data	\	3.77

Table 1: The average turns per dialogue of different datasets.

our self-constructed dialogues consistently under 4 turns, which demonstrates the efficiency.

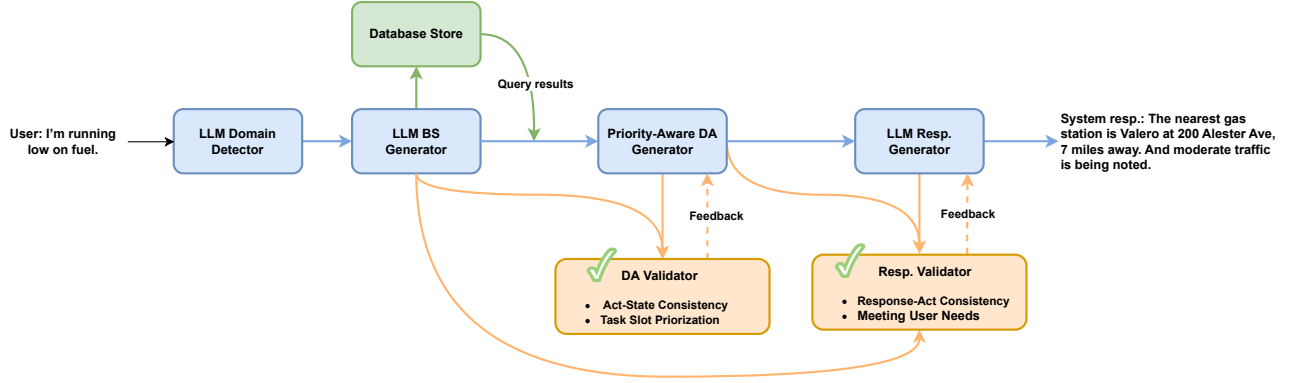


Figure 4: The CoVP Framework: Validators dynamically optimize dialogue acts and responses based on priority rules and driving context, with yellow dashed lines indicating iterative refinement loops.

4 Method

Recent advancements in LLMs have demonstrated their potential for end-to-end TOD systems [18, 19, 38]. While LLMs show promising conversational capabilities, studies reveal limitations in dialogue state tracking and consistency across stages [19], particularly in naturalness and efficiency for safety-critical scenarios like in-vehicle interactions.

To address these gaps, we propose **chain-of-validator prompting (CoVP)**—a framework combining priority-aware interaction design with chain validation to achieve both *efficiency* (reduced turns) and *naturalness* (human-aligned responses) (Figure 4). CoVP builds on prior work [19], adopting the belief state (BS) $\rightarrow$ dialogue acts (DA) $\rightarrow$ Response (Resp) pipeline. Our approach improves this process by integrating validators to reduce error propagation, ensuring consistency across BS, DA, and Resp, prioritizing task-specific slots, and effectively addressing user needs in dynamic contexts.

4.1 Validation Mechanism Design

Dialogue Act Validator. This validation component ensures the generated DAs strictly adhere to domain-specific task slot priority and maintain consistency with the current belief state. The validation process operates through three core mechanisms:

First, the validator ensures that DAs adhere to the priority rules by performing two checks: (1) blocking any DA containing level-2 (Lv2)/level-1 (Lv1) slot requests before all level-3 (Lv3) slots are confirmed, and (2) preventing DAs from combining slots of different priority levels. These checks are combined as follows:

$$\text{Validate}(DA) = \text{Reject if } (\exists s \in \mathcal{L}_3 \not\subseteq BS) \vee (IS_MIX(DA)), \quad (1)$$

where $\mathcal{L}_3$ denotes the mandatory slots defined in the priority lists. $IS_MIX(DA)$ means DAs mixing up with slots defined other levels.

Next, the validator performs act-state consistency checking.

$$\text{Validate}(DA) = \text{Reject if } BS(s) \neq \emptyset \text{ and } s \in \text{request}(DA), \quad (2)$$

where $BS(s)$ denotes the value of slot s in the belief state, and $\text{request}(DA)$ denotes the slots requested in the dialogue act.

Response Validator. This component ensures that linguistic responses strictly align with validated DAs and satisfy user needs as defined by the BS. It performs two levels of formal consistency checks:

First, the validator ensures that all slots specified in the validated DA are grounded in the response.

$$\text{Validate}(R) = \text{Reject if } \exists \sigma \in \Sigma(s), \sigma \not\subseteq R, \quad (3)$$

where σ denotes the slot mentioned in dialogue acts, $\Sigma(s)$ denotes the synonym set of slot s , and $\subseteq$ represents lexical or delexicalized containment.

Next, the validator ensures the response provides relevant information matching the user’s preferences or constraints specified in BS.

$$\text{Validate}(R) = \text{Reject if } \exists s \in BS \text{ and } s \not\subseteq R. \quad (4)$$

Feedback Loop. The system employs a bounded feedback loop where validation errors trigger regeneration with explicit error guidance. For DA failures, the validator appends feedback to the DA generator’s prompt; for response inconsistencies, it highlights omitted slots or conflicting values in the response generator’s context. Components allow up to 3 regeneration attempts, defaulting to the last valid result if retries exceed the limit.

4.2 Priority-aware Prompt Construction

In our approach, the construction of dialogue prompts is critical for ensuring that the system prioritizes relevant information at each stage of the interaction.

We intuitively embed priority-aware information directly into the DA. The prompt P_{da} combines domain-specific priority rules with history, user utterance, belief state, and feedback:

$$P_{da} = \text{DA_TEMPLATE}(H_{ctx}, U_{user}, BS_t, T_{rules}, F_{err}), \quad (5)$$

where T_{rules} represents the priority schema (e.g., Lv3 slots), H_{ctx} the conversation history, and F_{err} validation feedback from previous attempts. The template enforces the priority rules and provides contextual feedback to the LLM. A detailed dialogue act generation prompt is shown in Appendix A.2.

System	Inform	Success	BLEU	Comb.
Baseline	71.3	76.6	4.42	78.37
Full System	88.9	82.7	12.1	97.9

Table 2: Performance comparison between Baseline and Full System configurations.

For dialogue state tracking, a belief state prompt is shown in Appendix A.1. The prompt P_{bs} is constructed by integrating history and user utterance as follows:

$$P_{bs} = \text{BS_TEMPLATE}(H_{ctx}, U_{user}). \quad (6)$$

For response generation, the prompt P_{resp} integrates history, user utterance, dialogue act, and feedback:

$$P_{resp} = \text{RESP_TEMPLATE}(H_{ctx}, U_{user}, BS_t, DA_{valid}, F_{err}), \quad (7)$$

where DA_{valid} is the validated dialogue act and F_{err} is the validation feedback. A detailed response generation prompt is shown in Appendix A.3.

5 Objective Experiments

5.1 Experiment Setup

We evaluate our framework on our constructed in-vehicle dialogue dataset spanning 10 driving-centric scenarios. The dataset is partitioned into training (80%), validation (10%), and test (10%) sets, with scenario distributions preserved across splits through stratified sampling. Two system configurations are compared on the test set: (1) **Baseline** - a standard GPT-4o pipeline generating belief states (BS), dialogue acts (DA), and responses sequentially without constraints; (2) **Full System** - integrating priority-aware prompting and validation chains. Both systems employ identical GPT-4o (gpt-4o-2024-08-06) configurations and 3 regeneration attempts for validation failures.

Evaluation metrics focus on task effectiveness and linguistic quality: Inform Rate evaluates whether the system accurately learns user requests. Success Rate measures whether the generated DAs or responses contain all attributes in the requestable list. Language quality is evaluated via BLEU[28]. Combined Score balances both $0.5 \times (\text{Inform} + \text{Success}) + \text{BLEU}$.

5.2 Experiment Results and Analysis

As shown in Table 2, the Full System demonstrates significant improvements over the Baseline across all metrics, reflecting enhanced efficiency. The Full System achieves 88.9% Inform and 82.7% Success rates, compared to the Baseline’s 71.3% and 76.6%, respectively. This indicates that systematic enforcement of priority rules and validation checks reduces redundant interactions, improving task completion efficiency. In terms of naturalness, the Full System achieves a higher BLEU score (12.1 vs. 4.42), reflecting better linguistic quality in generated responses.

6 Subjective Experiments

To further evaluate the effectiveness of our proposed in-vehicle TOD, we designed and conducted subjective experiments focusing on five key dimensions: appropriateness, usability, efficiency, naturalness, and cognitive load.

6.1 Experiment Design

We use a within-subjects design where participants interact with two systems: a baseline instruction-based multi-turn system and our Full System using natural expressions based on our chain-of-validator prompting framework. Tasks cover 10 core in-vehicle scenarios from our dataset. Built with Alibaba ChatUI and React, the simulator includes ASR and TTS for realistic driver-system interaction. The simulator, displayed on an in-vehicle iPad, operates in a stationary car with a 240° curved screen simulating a dynamic environment. After a brief training, participants complete 10 dialogues (one per scenario) using both systems, followed by a questionnaire and a short interview after each session. We recruited 10 participants with valid driver’s licenses (5 males and 5 females), aged between 25 and 45, with an average driving experience of 4 years. All participants had prior experience using in-vehicle conversational systems.

6.2 Hypotheses

Our work aims to enhance the driving experience by reducing cognitive load and improving natural, efficient interactions in in-vehicle TOD systems. Prior research indicates that excessive dialogue turns or overly detailed instructions can increase distraction and frustration [6]. We hypothesize: **H1**. TOD systems minimizing unnecessary dialogue turns and prioritizing task-relevant information will be perceived as more efficient and useful, increasing user satisfaction. **H2**. The proposed system will be perceived as more natural, appropriate, and with a significantly lower cognitive load compared to traditional TOD. We evaluate appropriateness [27, 35], usability [44], naturalness [4], efficiency [32], and cognitive load [16] using 7-point Likert scales.

6.3 Results and Analysis

We compare the proposed system to the baseline across five dimensions: Appropriateness, Usability, Efficiency, Naturalness, and Cognitive Load.

Appropriateness. The proposed system scored higher ($M = 6.1$, $SD = 0.7$) than the baseline ($M = 4.3$, $SD = 1.0$; $t(9) = 5.89$, $p < 0.001$), with users finding it more suitable for in-vehicle use.

Usability. The proposed system outperformed the baseline ($M = 6.3$, $SD = 0.6$ vs. $M = 4.8$, $SD = 1.1$; $t(9) = 6.45$, $p < 0.001$), with participants praising its intuitive interface.

Efficiency. The proposed system scored higher ($M = 6.4$, $SD = 0.5$ vs. $M = 4.7$, $SD = 1.2$; $t(9) = 7.12$, $p < 0.001$), aligning with **H1** by reducing dialogue turns and task completion time.

Naturalness. The proposed system was rated as more natural ($M = 6.2$, $SD = 0.8$ vs. $M = 4.5$, $SD = 1.0$; $t(9) = 6.78$, $p < 0.001$), supporting **H2** with its conversational flexibility.

Cognitive Load. The proposed system significantly reduced cognitive load ($M = 3.5$, $SD = 1.1$ vs. $M = 5.2$, $SD = 1.3$; $t(9) = 5.34$, $p < 0.001$), further validating **H2**.

90% of participants preferred the proposed system, citing its efficiency, naturalness, and lower cognitive load. In the post-experiment interviews, participants provided additional insights into their experiences: Many appreciated the Full System’s ability to adapt to their natural language inputs, reducing the need for precise phrasing. Several participants noted that the baseline system often felt “robotic” and “frustrating”, especially when it failed to understand their requests or required repetitive clarifications. Some participants suggested further improvements, such as integrating more contextual awareness (e.g., recognizing the driver’s current activity or location) to enhance the system’s responsiveness.

All these results strongly support **H1** and **H2**, demonstrating the proposed system’s superiority in appropriateness, usability, efficiency, naturalness, and cognitive load. These findings validate the effectiveness of the chain-of-validator prompting framework for optimizing in-vehicle TOD systems.

7 Limitations and Future Work

Our study has several limitations. First, the small sample size ($N = 10$) may limit the generalizability of our findings. Future work should involve larger-scale studies to better capture diverse user experiences. Second, the controlled experimental environment, while simulating driving scenarios, may not fully reflect real-world driving conditions. Validating the framework in actual driving settings is essential for future research. Finally, while our dataset provides valuable guidance and labeled data for prompt engineering with large language models, its limited scale constrains its use for training or fine-tuning larger models. Expanding the dataset to include more diverse and comprehensive in-vehicle interactions will enhance model generalization and improve performance in real-world applications.

8 Conclusion

We mitigate the challenges in traditional in-vehicle TOD systems, which require detailed, multi-turn instructions, increasing cognitive load and distraction. Our human-centered approach emphasizes natural interactions and efficiency by introducing a novel dataset for 10 in-vehicle scenarios and a chain-of-validator prompting framework that validates responses in real time, reducing unnecessary turns. Experiments show our method outperforms the conventional prompting approach in task completion, with subjective evaluations highlighting superior appropriateness, usability, efficiency, naturalness, and reduced cognitive load, enhancing user satisfaction and safety.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (U23B2057, 62176185), and the Youth Fund Special Project at Hangzhou Dianzi University Information Engineering College (KYP0323010).

References

- [1] Jacob Andreas, John Bufe, David Burkett, Charles Chen, Josh Clausman, Jean Crawford, Kate Crim, Jordan DeLoach, Leah Dörner, Jason Eisner, Hao Fang, Alan Guo, David Hall, Kristin Hayes, Kellie Hill, Diana Ho, Wendy Iwaszuk, Smriti Jha, Dan Klein, Jayant Krishnamurthy, Theo Lanman, Percy Liang, Christopher H. Lin, Ilya Lintsakh, Andy McGovern, Aleksandr Nisnevich, Adam Pauls, Dmitriy Petters, Brent Read, Dan Roth, Subho Roy, Jesse Rusak, Beth Short, Div Slomin, Ben Snyder, Stephon Striplin, Yu Su, Zachary Tellman, Sam Thomson, Andrei Vorobev, Izabela Witoszko, Jason Wolfe, Abby Wray, Yuchen Zhang, and Alexander Zotov. 2020. Task-Oriented Dialogue as Dataflow Synthesis. *Transactions of the Association for Computational Linguistics* 8 (2020), 556–571. doi:10.1162/tacl_a_00333
- [2] Paweł Budzianowski, Tsung-Hsien Wen, Bo-Hsiang Tseng, Inigo Casanueva, Pei-Hao Su, Stefan Ultes, and Steve Young. 2018. MultiWOZ - A Large-Scale Multi-Domain Wizard-of-Oz Dataset for Task-Oriented Dialogue Modelling. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- [3] Bill Byrne, Karthik Krishnamoorthi, Chinnadhurai Sankar, Arvind Neelakantan, Ben Goodrich, Daniel Duckworth, Semih Yavuz, Amit Dubey, Kyu-Young Kim, and Andy Cedilnik. 2019. Taskmaster-1: Toward a Realistic and Diverse Dialog Dataset. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (Eds.). Association for Computational Linguistics, Hong Kong, China, 4516–4525. doi:10.18653/v1/D19-1459
- [4] Jianqin CAO, Jingyu ZHANG, Liang ZHANG, and Xiaoyu WANG. 2023. The psychological structure and influence of interactive naturalness. *Acta Psychologica Sinica* 55, 1 (2023), 55.
- [5] S Devitt. 2018. Trustworthiness of autonomous systems. *Foundations of trusted autonomy (Studies in Systems, Decision and Control, Volume 117)* (2018), 161–184.
- [6] Huifang Du, Xuejing Feng, Jun Ma, Meng Wang, Shiyu Tao, Yijie Zhong, Yuan-Fang Li, and Haofen Wang. 2024. Towards Proactive Interactions for In-Vehicle Conversational Assistants Utilizing Large Language Models. *arXiv preprint arXiv:2403.09135* (2024).
- [7] Huifang Du, Shuqin Li, Minghao Wu, Xuejing Feng, Yuan-Fang Li, and Haofen Wang. 2024. Rewarding What Matters: Step-by-Step Reinforcement Learning for Task-Oriented Dialogue. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 8030–8046. doi:10.18653/v1/2024.findings-emnlp.472
- [8] Huifang Du, Shiyu Tao, Xuejing Feng, Jun Ma, and Haofen Wang. 2024. From Passive to Active: Towards Conversational In-Vehicle Navigation Through Large Language Models. In *International Conference on Human-Computer Interaction*. Springer, 159–172.
- [9] Mihail Eric, Lakshmi Krishnan, Francois Charette, and Christopher D Manning. 2017. Key-Value Retrieval Networks for Task-Oriented Dialogue. In *Proceedings of the 18th Annual SIGdial Meeting on Discourse and Dialogue*. 37–49.
- [10] Mihail Eric, Lakshmi Krishnan, Francois Charette, and Christopher D. Manning. 2017. Key-Value Retrieval Networks for Task-Oriented Dialogue. In *Proceedings of the 18th Annual SIGdial Meeting on Discourse and Dialogue*, Kristiina Jokinen, Manfred Stede, David DeVault, and Annie Louis (Eds.). Association for Computational Linguistics, Saarbrücken, Germany, 37–49. doi:10.18653/v1/W17-5506
- [11] Raul Fernandez-Rojas, Anthony Perry, Hemant Singh, Benjamin Campbell, Saber Elsayed, Robert Hunjet, and Hussein A Abbass. 2019. Contextual awareness in human-advanced-vehicle systems: a survey. *IEEE Access* 7 (2019), 33304–33328.
- [12] Kristina Lundholm Fors and Jessica Villing. 2011. Reducing Cognitive Load in In-Vehicle Dialogue System Interaction. In *Proceedings of the 15th Workshop on the Semantics and Pragmatics of Dialogue*. SemDial, Los Angeles, California, 55–62. Vinnova funded DICO project.
- [13] Jianfeng Gao, Michel Galley, and Lihong Li. 2019. Neural approaches to conversational AI. *Foundations and Trends in Information Retrieval* 13, 2-3 (2019), 127–298.
- [14] Maria Guinea, Isabel Litton, Rigel Smirolodo, Irma Nitsche, and Eric Sax. 2021. A Proactive Context-Aware Recommender System for In-Vehicle Use. In *Proceedings of the 2020 4th International Conference on Vision, Image and Signal Processing (Bangkok, Thailand) (ICVISIP 2020)*. Association for Computing Machinery, New York, NY, USA, Article 32, 8 pages. doi:10.1145/3448823.3448852
- [15] Aman Gupta, Anirudh Ravichandran, Ziji Zhang, Swair Shah, Anurag Beniwal, and Narayanan Sadagopan. 2024. DARD: A multi-agent approach for task-oriented dialog systems. *arXiv preprint arXiv:2411.00427* (2024).
- [16] SG Hart. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. *Human mental workload/Elsevier* (1988).
- [17] Ehsan Hosseini-Asl, Bryan McCann, Chien-Sheng Wu, Semih Yavuz, and Richard Socher. 2020. A simple language model for task-oriented dialogue. *Advances in Neural Information Processing Systems* 33 (2020), 20179–20191.
- [18] Zhiyuan Hu, Yue Feng, Yang Deng, Zekun Li, See-Kiong Ng, Anh Tuan Luu, and Bryan Hooi. 2023. Enhancing Large Language Model Induced Task-Oriented Dialogue Systems Through Look-Forward Motivated Goals. *arXiv:2309.08949 [cs.CL]* <https://arxiv.org/abs/2309.08949>
- [19] Vojtěch Hudeček and Ondřej Dusek. 2023. Are Large Language Models All You Need for Task-Oriented Dialogue?. In *Proceedings of the 24th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, Svetlana Stoyanchev, Shafiq Joty, David Schlangen, Ondřej Dusek, Casey Kennington, and Malihe Alikhani (Eds.). Association for Computational Linguistics, Prague, Czechia, 216–228.

- doi:10.18653/v1/2023.sigldl-1.21
- [20] Youngsoo Jang, Jongmin Lee, and Kee-Eung Kim. 2022. GPT-critic: Offline reinforcement learning for end-to-end task-oriented dialogue systems. In *International Conference on Learning Representations*.
- [21] S. Lee, H. Schulz, A. Atkinson, J. Gao, K. Suleman, L. El Asri, M. Adada, M. Huang, S. Sharma, W. Tay, and X. Li. 2019. Multi-domain Task-completion Dialog Challenge. In *Dialog System Technology Challenges* 8.
- [22] Zekun Li, Baolin Peng, Pengcheng He, Michel Galley, Jianfeng Gao, and Xifeng Yan. 2024. Guiding large language models via directional stimulus prompting. *Advances in Neural Information Processing Systems* 36 (2024).
- [23] Zhaohang Lin, Andrea Madotto, Genta Indra Winata, and Pascale Fung. 2020. MinTL: Minimalist Transfer Learning for Task-Oriented Dialogue Systems. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, Online, 3391–3405. doi:10.18653/v1/2020.emnlp-main.273
- [24] Shikib Mehri, Mihail Eric, and Bo-Hsiang Tseng. 2020. DialogGLUE: A Natural Language Understanding Benchmark for Task-Oriented Dialogue. *arXiv preprint arXiv:2009.13570* (2020).
- [25] Baolin Peng, Chunyuan Li, Jinchao Li, Shahin Shayandeh, Lars Liden, and Jianfeng Gao. 2021. Soloist: Building task bots at scale with transfer learning and machine teaching. *Transactions of the Association for Computational Linguistics* 9 (2021), 807–824.
- [26] Baolin Peng, Chenguang Zhu, Chunyuan Li, Xiujun Li, Jinchao Li, Michael Zeng, and Jianfeng Gao. 2020. Few-shot Natural Language Generation for Task-Oriented Dialog. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, Online, 172–182. doi:10.18653/v1/2020.findings-emnlp.17
- [27] Zhenhui Peng, Yunhwan Kwon, Jiaan Lu, Ziming Wu, and Xiaojuan Ma. 2019. Design and evaluation of service robot’s proactivity in decision-making support process. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [28] Matt Post. 2018. A Call for Clarity in Reporting BLEU Scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, Ondřej Bojar, Rajen Chatterjee, Christian Federmann, Mark Fishel, Yvette Graham, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, Christof Monz, Matteo Negri, Aurélie Névoul, Mariana Neves, Matt Post, Lucia Specia, Marco Turchi, and Karin Verspoor (Eds.). Association for Computational Linguistics, Brussels, Belgium, 186–191. doi:10.18653/v1/W18-6319
- [29] Jing Qian, Li Dong, Yelong Shen, Furu Wei, and Weizhu Chen. 2022. Controllable Natural Language Generation with Contrastive Prefixes. In *Findings of the Association for Computational Linguistics: ACL 2022*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 2912–2924. doi:10.18653/v1/2022.findings-acl.229
- [30] Govardana Sachithanandam Ramachandran, Kazuma Hashimoto, and Caiming Xiong. 2022. [CASPI] Causal-aware Safe Policy Improvement for Task-oriented Dialogue. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 92–102. doi:10.18653/v1/2022.acl-long.8
- [31] Abhinav Rastogi, Songhao Zang, Saurav Sunkara, Raghav Gupta, and Sunayana Khaitan. 2020. Towards Scalable Multi-Domain Conversational Agents: The Schema-Guided Dialogue Dataset. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [32] Clemencia Siro, Mohammad Aliannejadi, and Maarten de Rijke. 2022. Understanding user satisfaction with task-oriented dialogue systems. In *Proceedings of the 45th International ACM SIGIR conference on research and development in information retrieval*. 2018–2023.
- [33] Clemencia Siro, Mohammad Aliannejadi, and Maarten De Rijke. 2023. Understanding and predicting user satisfaction with conversational recommender systems. *ACM Transactions on Information Systems* 42, 2 (2023), 1–37.
- [34] Zhe Song, Ye Yang, and Mengying Deng. 2023. Trust Repair of Automated Driving System: A New In-Vehicle Communication Strategy of Voice Assistant. In *International Conference on Human-Computer Interaction*. Springer, 99–106.
- [35] Cristen Torrey, Susan R Fussell, and Sara Kiesler. 2013. How a robot should give advice. In *2013 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 275–282.
- [36] Joseph Weizenbaum. 1966. ELIZA—a computer program for the study of natural language communication between man and machine. *Commun. ACM* 9, 1 (1966), 36–45.
- [37] Wenquan Wu, Zhen Guo, Xiangyang Zhou, Hua Wu, Xiyuan Zhang, Rongzhong Lian, and Haifeng Wang. 2019. Proactive Human-Machine Conversation with Explicit Conversation Goals. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, Florence, Italy, 3794–3804. <https://aclanthology.org/P19-1369.pdf>
- [38] Heng-Da Xu, Xian-Ling Mao, Puhai Yang, Fanshu Sun, and Heyan Huang. 2024. Rethinking Task-Oriented Dialogue Systems: From Complex Modularity to Zero-Shot Autonomous Agent. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 2748–2763. doi:10.18653/v1/2024.acl-long.152
- [39] Heng-Da Xu, Xian-Ling Mao, Puhai Yang, Fanshu Sun, and He-Yan Huang. 2024. Rethinking Task-Oriented Dialogue Systems: From Complex Modularity to Zero-Shot Autonomous Agent. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2748–2763.
- [40] Yunyi Yang, Yunhao Li, and Xiaojun Quan. 2021. Ubar: Towards fully end-to-end task-oriented dialog system with gpt-2. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 14230–14238.
- [41] Steve Young, Milica Gašić, Blaise Thomson, and Jason D Williams. 2013. Pomdp-based statistical spoken dialog systems: A review. *Proc. IEEE* 101, 5 (2013), 1160–1179.
- [42] Xiao Yu, Maximilian Chen, and Zhou Yu. 2023. Prompt-Based Monte-Carlo Tree Search for Goal-oriented Dialogue Policy Planning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 7101–7125. doi:10.18653/v1/2023.emnlp-main.439
- [43] Zheng Zhang, Ryuichi Takanobu, Qi Zhu, MinLie Huang, and XiaoYan Zhu. 2020. Recent advances and challenges in task-oriented dialog systems. *Science China Technological Sciences* 63, 10 (2020), 2011–2027.
- [44] Dilawar Shah Zwakman, Debajyoti Pal, Tuul Triyason, and Chonlameth Arpanikondt. 2020. Voice usability scale: measuring the user experience with voice assistants. In *2020 IEEE International Symposium on Smart Electronic Systems (iSES)(Formerly iNiS)*. IEEE, 308–311.

A Prompt Construction Examples

A.1 Belief State Prompt

As shown in Table 3, the belief state prompt captures entity values from the last utterance of the conversation according to examples.

A.2 Priority-Aware Dialogue Act Prompt

As shown in Table 4, the dialogue act prompt embeds priority rules into the dialogue act generation process. It contains a domain-specific priority schema and action selection guidelines.

A.3 Response Prompt

As shown in Table 5, the response prompt generates a natural response based on the belief state and dialogue acts. It follows a logical order of actions and uses natural language to express the dialogue acts.

Capture entity values from last utterance of the conversation according to examples.
Focus only on the values mentioned in the last utterance.
Capture pair "entity:value" separated by colon and no spaces in between.
Separate entity:value pairs by commas. Like this: entity1:'value1',entity2:'value2',entity3:'value3'
Values that should be captured are:

- "pricerange" that specifies the price range of the restaurant (cheap/moderate/expensive)
- "area" that specifies the area where the restaurant is located (north/east/west/south/centre)
- "food" that specifies the type of food the restaurant serves
- "name" that specifies the name of the restaurant
- "bookday" that specifies the day of the booking
- "booktime" that specifies the time of the booking
- "bookpeople" that specifies for how many people is the booking made

Do not capture any other values!
If not specified, leave the value empty.
Now complete the following example:
input: history
Customer: user_utterance
output:
state:

Table 3: The belief state prompt for the “restaurant” domain, adapted from [19].

You are a dialogue action planner for a restaurant booking system. Follow these guidelines:

Action Types (in priority order):

1. request - Ask for missing required information
2. inform - Provide information about restaurants
3. recommend - Suggest specific restaurants
4. book - Make a reservation
5. confirm - Confirm booking details
6. general - General responses (welcome, goodbye, etc.)

Format:

[domain_name] [action_type] slot_name ... [action_type] slot_name ... [action_type] slot_name ...

For example: [restaurant] [inform] phone [general] [reqmore]

Priority Guidelines:

The priority levels help determine which information to request or provide first:

- Always handle Level 3 slots before moving to Level 2 or Level 1
- Only request Level 2 & 1 slots if contextually relevant
- Combine requests for slots of the same priority level

For find_restaurant intent:

Level 3 (Must collect all):

- name (if searching specific one)
- area
- food
- address

Level 2 (Ask after Level 3):

- pricerange

Level 1 (Provide only if requested):

- phone
- postcode

For book_restaurant intent:

Level 3 (Must collect all):

- name
- bookday
- booktime

Level 2 (Ask after Level 3):

- pricerange
- bookpeople
- food

Level 1 (Provide only if requested):

- phone
- postcode
- address

Action Selection Guidelines:

1. For finding restaurants:
 - First REQUEST any missing Level 3 slots
 - Then INFORM/RECOMMEND based on available information
 - Only REQUEST Level 2/1 slots if user shows interest
2. For booking:
 - Must have all Level 3 slots before BOOK action
 - REQUEST missing Level 3 slots first
 - Then collect necessary Level 2 information
 - Only discuss Level 1 slots if requested

Now complete the following example:

history:

user utterance:

current state:

database results:

feedback:

output:

dialogue acts:

Table 4: The dialogue act prompt for the “restaurant” domain, embedded with priority rules.

You are a hotel booking assistant.
Generate a natural response based on the belief state and dialogue acts.
Follow these guidelines:

1. Be concise but polite
2. Address all actions in a logical order:
 - REQUEST before INFORM
 - INFORM before BOOK
 - BOOK before CONFIRM
3. Use natural language to express the dialogue acts

Now complete the following example:

history:
user utterance:
state:
dialogue acts:
database results:
output:
response:

Table 5: The response prompt for “restraurant” domain.